TorqBall — Stretch Modes

2026-09-06. The four stretch modes named in [[TorqBall — Design Study v1]] and pointed to from [[TorqBall — Rulebook]] §5 — Hoops, Puck, Rumble, and a symmetric Hazard/Arena Variant. None are required for a v1 launch (v1 = Standard + Duel + Big Party, per the rulebook). Each entry below states what the mode is, the exact build + test that unlocks it, a rough effort estimate, and a go/no-go bar — so each can be picked up independently once the base game (and [[TorqBall — Ball & Boards Test]]) has proven out.

---

1. Hoops

What it is: the goal comes down to car height via a ramp — a raised net/ramp-basket instead of a ground-level opening. Push the ball up a wedge into a basket/bin at ~25–40 cm, skee-ball logic. The Scoop boost ([[TorqBall — Boost & Attachments]]) is built to feed this exact shot — lifting the ball a little as you approach the ramp.

What we need to build + the test that unlocks it:

- **Build:** 2 DIY ramp-basket goals — a low wedge (plywood or foam-core board) rising from pitch-floor height up to a net/bin lip at ~25–40 cm, positioned where a standard goal opening would sit. Ramp angle needs to be shallow enough for a speed-capped car to shove the ball up it, not so shallow that the ball just rolls back down before reaching the lip.
- **Test — the roll-back mini-test:** push the winning ball (from [[TorqBall — Ball & Boards Test]]) up 2–3 candidate ramp angles with a capped car. Does the ball reach the basket and stay (score), or roll back down before the lip (fail)? Try with and without the Scoop attachment to confirm it meaningfully helps. ~10–15 pushes per ramp angle, log make/roll-back/wedge for each.

Rough effort: small-medium — 2 ramp-basket units to build (plywood/foam-core, a bin or net top), one afternoon of angle-testing. Comparable in scale to one TorqSiege weapon-print run.

Go/no-go bar:

- **GO** = at least one ramp angle scores a make (ball reaches and stays in the basket) on a majority of pushes (≥60%) at capped speed, with the Scoop attachment measurably improving the make rate. → Build 2 units for the event, add Hoops to the playlist as an occasional heat.
- **NO-GO** = the ball reliably rolls back at every safe ramp angle, or only works at speeds above the cap. → Shelve Hoops; the ramp geometry needs rework (a shorter/steeper ramp with a mechanical assist, or a different ball) before it's worth a full build.

---

2. Puck

What it is: swap the ball for a big sliding disc (a "Snow Day"-style puck) — a passier, faster, differently-weighted game. The cheapest variant to try since it reuses the same boarded pitch and goals, just a different play object.

What we need to build + the test that unlocks it:

- **Build:** a large sliding disc — light, low-friction underside, big enough to read on camera (similar diameter band to the winning ball, ~40–60 cm), foam or lightweight plastic construction so a car-to-puck hit is safe.
- **Test — the puck-on-turf slide test:** push the disc across the actual pitch surface (turf) at capped car speed and measure how far/cleanly it slides. Astroturf has real friction, so this may fail on turf even though it works on a smooth floor.
- **Turf vs bare-floor patch decision:** if the slide test fails on turf (the disc drags and dies rather than sliding), the two options are (a) a slicker underside on the disc (a smooth base layer), or (b) a small bare-floor or laminate patch laid over the turf inside the boarded pitch just for Puck sessions. Test both before concluding the mode needs a floor change.

Rough effort: small — one disc to build/buy, a single afternoon slide test, only a floor-patch cost if the turf option fails. The cheapest of the four stretch modes.

Go/no-go bar:

- **GO** = the disc slides cleanly and controllably across the actual pitch surface (turf, or turf + a patch) at capped speed — reads as genuinely different from ball-play, not just a slower, stickier version of the same game. → Add Puck as a cheap between-round variant.
- **NO-GO** = the disc drags to a stop almost immediately on turf and even a floor patch doesn't fix it economically. → Shelve Puck; not worth a floor-patch spend for a marginal variant.

---

3. Rumble

What it is: the boost system ([[TorqBall — Boost & Attachments]]) turned to 11 — boost pads everywhere, every boost active, maximum chaos. Not a separate build — it's TorqBall with the existing boost kit dialled up, run as its own high-energy mode once the base boost system exists.

What we need to build + the test that unlocks it: nothing new. Rumble is **free once the boost set exists** — it's a tuning/programming exercise (how many pads, how densely packed, which boosts are in rotation), not a build. The only "test" is a playtest of the pad density and boost mix to find a chaos level that's fun rather than unplayable (e.g. does 6+ simultaneous Plow/Grabber cars on one small ball just jam the pitch?).

Rough effort: minimal — a tuning pass on pad count/placement and boost mix, once [[TorqBall — Boost & Attachments]] is built and working in Big Party sessions.

Go/no-go bar:

- **GO** = a playtest with pads turned up finds a density/mix where the pitch still resolves into visible shoves and scores rather than a permanent gridlock. → Lock the pad layout for Rumble, add it to the playlist as a variety round.
- **NO-GO** = every density tried either jams the pitch solid or is functionally indistinguishable from a boost-off Big Party. → Keep boosts at the lighter Big Party density; don't stand up Rumble as its own labelled mode yet.

---

4. Hazard / Arena Variant

What it is: a symmetric, announced terrain variant — a slick centre patch, a boost lane, or a slow-mat zone built into the pitch, identical and fair for both teams. **Not random per-possession traps** — those belong in TorqFall's chaos gauntlet, not here (locked decision, [[TorqBall — Design Study v1]]). This variant changes the arena for a whole match, announced up front, same for both sides.

What we need to build + the test that unlocks it:

- **Build:** a dry, safe terrain patch inside the boarded pitch — a **low-friction vinyl patch** (the "ice/water slick" stand-in) or a **high-friction mat** (the "mud/slow" stand-in), laid as a standing arena feature for a full match. *(Note: this is the variant's own low-friction feature, costed here — it is not shared with [[TorqBall — Boost & Attachments]]'s Turbo boost, which dropped its "lane" mechanism entirely on the 2026-09-06 fix: a low-friction surface reduces traction rather than adding speed, so Turbo is now solely a fleet-wide throttle toggle. This Hazard Variant patch is honest about what low friction actually does — a slide/chaos effect for a full symmetric match, not a speed mechanic.)*
- **Test:** lay the patch in a fixed, symmetric position (e.g. dead centre, or mirrored left/right of the centre line) and run a full test match on it. Confirm: (a) the patch is genuinely fair — neither team gets a positional advantage from where it sits, (b) it changes gameplay in a visible, fun way (a car sliding through the centre, or bogging down in a slow zone) without becoming a hazard to safety (a car losing control into the boards or a spectator-adjacent panel).
- **Explicitly out of scope:** any marshal-triggered pop-up bump or random per-possession trap — that mechanic is reserved for TorqFall, not TorqBall, per the locked decision.

Rough effort: small — a simple vinyl slick patch or high-friction mat, ~2–3 m², ~Rs 1,000–3,000 (retail vinyl sheeting or a rubber-backed mat, cut to size; same unit pricing as a low-friction mat elsewhere in the doc set), one test match to confirm fairness and safety. No new hardware category, and no dependency on the Turbo boost (which no longer uses a lane — see the note above).

Go/no-go bar:

- **GO** = the test match shows a visibly different, fair, safe game (both teams affected equally, no loss-of-control incident). → Add the Hazard/Arena Variant as an announced special-round option in the tournament programme (e.g. one bracket round runs "Slick Centre TorqBall").
- **NO-GO** = the patch causes a safety issue (cars losing control near the boards/spectator line) or reads as unfair in practice despite symmetric placement. → Drop that specific patch; try a milder friction differential or a smaller patch footprint before retesting.

Summary — build order if picking these up

Mode	Effort	Depends on	Best next step
Rumble	Minimal (tuning only)	[[TorqBall — Boost & Attachments]] built and working	Run a pad-density playtest during a Big Party session
Puck	Small	[[TorqBall — Ball & Boards Test]] boards passed (pitch already exists)	Slide-test one disc on the actual turf

Mode	Effort	Depends on	Best next step
Hazard/Arena Variant	Small	A simple vinyl/mat patch (its own cost, ~₹1,000–3,000 — not shared with Turbo, which dropped its lane mechanism)	One symmetric test match, checked for fairness + safety
Hoops	Small–medium	Ball locked (Part A of Ball & Boards Test), Scoop boost built	Build 2 ramp units, run the roll-back angle test

None of these block a v1 launch. Pick them up in roughly the order above — Rumble and the Hazard Variant are nearly free once the core kit exists; Puck is a cheap standalone experiment; Hoops is the only one needing a real dedicated build (2 ramp-basket units) and should wait until Standard/Duel/Big Party have proven out on the boarded pitch.

Verified + changelog (2026-09-06): *first build, covering all four stretch modes named in [[TorqBall — Design Study v1]] and pointed to from [[TorqBall — Rulebook]] §5. For each mode (Hoops, Puck, Rumble, Hazard/Arena Variant): stated what it is, the exact build, the specific test that gates it, a rough effort estimate, and an explicit GO/NO-GO bar. Hoops specced with a roll-back ramp-angle test tied to the Scoop boost; Puck specced with a turf-slide test and a turf-vs-bare-floor-patch decision; Rumble stated as free/tuning-only once the boost system exists; Hazard/Arena Variant scoped strictly to symmetric, announced terrain (no random per-possession traps — that stays reserved for TorqFall per the locked decision). Closed with a build-order summary table so any one mode can be picked up independently without blocking the v1 launch (Standard + Duel + Big Party).*

Second pass (2026-09-06, examiner fix): *decoupled the Hazard/Arena Variant's low-friction patch from the Turbo boost's "lane," since [[TorqBall — Boost & Attachments]] §2 dropped that lane mechanism entirely (a low-friction surface reduces traction, it doesn't add speed — Turbo is now solely a fleet-wide throttle toggle). Gave the variant its own patch cost (~Rs 1,000–3,000) instead of pointing at a Turbo line that no longer exists.*